Parameter Estimation for Linear Regression Type Model Structures – Case Study

(具有线性回归类型模型结构的参数估计 – 案例分析)

中国大学MOOC

Ex.2 LSE under Effect of Model Structure Error

- **Consider the following system** (考虑以下系统)

$$y(t) = b_0 + b_1 u(t) + b_2 u(t)^2 + e(t)$$
$$= \varphi_a(t)^T \theta_a + e(t)$$

$$e(t) = \sigma.\text{randn}$$
$$\varphi_a(t)^T = [1 \quad u(t) \quad u^2(t)]$$
$$\theta_a = [b_0 \quad b_1 \quad b_2]^T$$

where $e(t)$ is a Gaussian distributed zero mean white noise sequence with a standard deviation of σ (***e*(*t*)** 是高斯分布零均值白噪声序列，标准差是**σ**).

We would like to use the following four sets of different model structures to fit the actual data respectively: All four model structures are linear with respect to their model parameters (用以下四种不同的模型结构来拟合数据：所有的四种模型结构是关于模型参数线性的).

- **LSE algorithm to obtain model parameters** (获得模型参数的最小二乘算法)

$$\hat{\theta}_M(t) = P_M(t)[\sum_{i=1}^{t} \varphi_M(i) y(i)], \qquad P_M(t) = [\sum_{i=1}^{t} \varphi_M(i) \varphi_M(i)^T]^{-1}$$

where M = I, II, III, IV for each model respectively.

Ex.2 LSE under Effect of Model Structure Error

- **Analysis of Different Model Structure and Its Errors** (不同模型结构和误差分析)

	Assumed Model Structure (假设的模型结构)	Regrsessor and Model Parameters (回归量和模型参数)
M1	$\hat{y}(t,\theta_I)=c_0$ $=\varphi_I(t)\theta_I$	$\varphi_I(t)=1$ $\theta_I=c_0$
M2	$\hat{y}(t,\theta_{II})=c_0+c_1u(t)$ $=\varphi_{II}(t)^T\theta_{II}$	$\varphi_{II}(t)=[1,u(t)]^T$ $\theta_{II}=[c_0,c_1]^T$
M3	$\hat{y}(t,\theta_{III})=c_0+c_1u(t)+c_2u(t)^2$ $=\varphi_{III}(t)^T\theta_{III}$	$\varphi_{III}(t)=[1,u(t),u(t)^2]^T$ $\theta_{III}=[c_0,c_1,c_2]^T$
M4	$\hat{y}(t,\theta_{IV})=c_0+c_1u(t)+c_2u(t)^2+c_3u(t)^3$ $=\varphi_{IV}(t)^T\theta_{IV}$	$\varphi_{IV}(t)=[1,u(t),u(t)^2,u(t)^3]^T$ $\theta_{IV}=[c_0,c_1,c_2,c_3]^T$

Table 1 Model structures (模型结构)

	Model Error (模型误差)	Uncorrelated with Regressor? (与回归量不相关?)
M1	$\varepsilon_\theta(t)=b_1u(t)+b_2u(t)^2+e(t)$ with $\theta_I^0=b_0$	Neither white nor uncorrelated with $\varphi_I(t)$ in general (既非白噪声也非与$\varphi_I(t)$不相关)
M2	$\varepsilon_\theta(t)=b_2u(t)^2+e(t)$ with $\theta_{II}^0=[b_0,b_1]^T$	Neither white nor uncorrelated with $\varphi_{II}(t)$ in general (既非白噪声也非与$\varphi_{II}(t)$不相关)
M3	$\varepsilon_\theta(t)=e(t)$ with $\theta_{III}^0=[b_0,b_1,b_2]^T$	white and uncorrelated with $\varphi_{III}(t)$ (是白噪声且与$\varphi_{III}(t)$不相关)
M4	$\varepsilon_\theta(t)=e(t)$ with $\theta_{IV}^0=[b_0,b_1,b_2,0]^T$	white and uncorrelated with $\varphi_{IV}(t)$ (是白噪声且与$\varphi_{IV}(t)$不相关)

Table 2 Model structure errors (模型结构误差)

Ex.2 LSE under Effect of Model Structure Error

- **LSE Results under Zero Random Noise** (零随机噪声下的最小二乘估计结果)

Model	$\hat{b}_0(t)$	$\hat{b}_1(t)$	$\hat{b}_2(t)$	$\hat{b}_3(t)$	$J(t, \hat{\boldsymbol{\theta}}(t))$
M1	2.6				14.56
M2	0.5	0.7			0.84
M3	1.0	0.1	0.1		0
M4	1.0	0.1	0.1	0	0

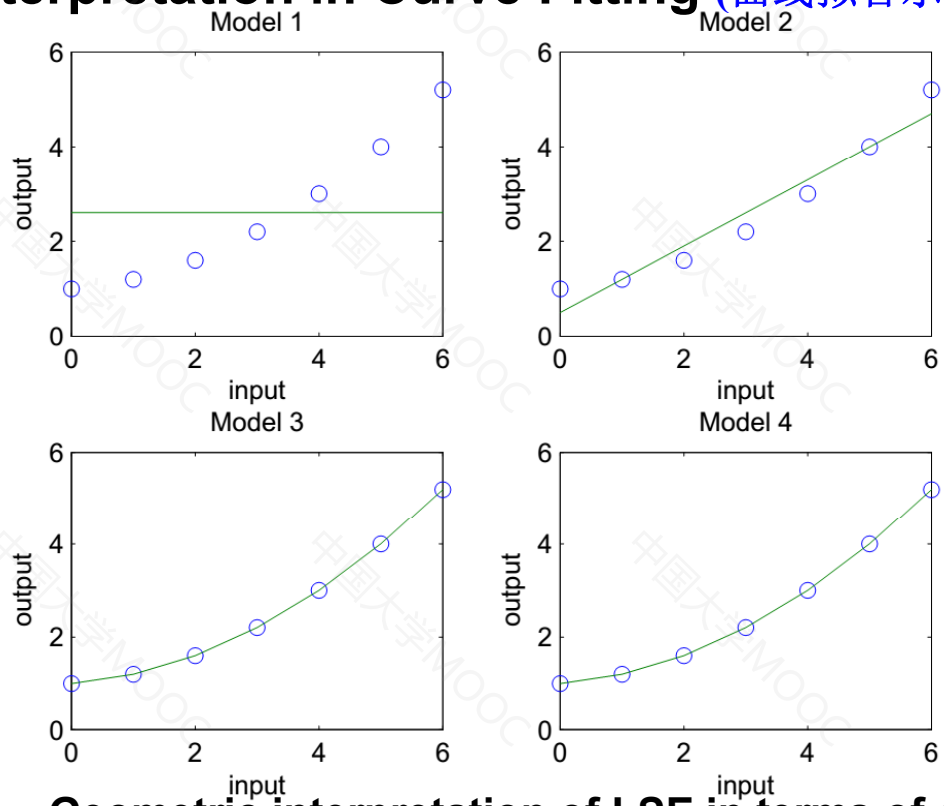
Table 3. LSE results in absence of random noises ($\sigma = 0$)

Actual system with $b_0 = 1$, $b_1 = 0.1$, $b_2 = 0.1$, and $\sigma = 0$

when the actual system is absence of random noise $e(t)$, the LSE is expected to exactly recover the true model parameters $\boldsymbol{\theta}^o_{III}$ and $\boldsymbol{\theta}^o_{IV}$ for the model structure M3 and M4 respectively, as long as experimental conditions are controlled so that $P_{III}(t)$ and $P_{IV}(t)$ are finite for the collected experimental data set (当实际系统没有随机噪声时,只要实验条件受控, 对所收集的数据集来说$P_{III}(t)$ 和 $P_{IV}(t)$是有限的, 最小二乘估计能分别精确复现M3和M4模型结构下真实的模型参数)

Ex.2 LSE under Effect of Model Structure Error

- **Interpretation in Curve Fitting** (曲线拟合示意)



(i) the assumed linear regression form (the solid lines in the figures) (实线表示所假设的线性回归形式)

(ii) the measured data set (the dots in the figures) (点表示实测数据)

Geometric interpretation of LSE in terms of curve fitting

Ex.2 LSE under Effect of Model Structure Error

- **LSE Results under Non-Zero Random Noise** (非零随机噪声下的最小二乘估计结果)

Actual system with $b_0 = 1$, $b_1 = 0.1$, $b_2 = 0.1$, and $\sigma = 0.1$,

Small data set: the actual output is generated using an input sequence of $u(i) = i-1$ for $i = 1, \ldots, t = 7$ only (采样数据集是由输入序列产生的真实输出)

Table 4. LSE results for M3 with random noises ($\sigma = 0.1$) and $t = 7$

Run	$\hat{b}_0(t)$	$\hat{b}_1(t)$	$\hat{b}_2(t)$	$\hat{b}_3(t)$	$J(t, \hat{\boldsymbol{\theta}}(t))/t$
1	1.01	0.11	0.097		0.0025
2	0.902	0.236	0.077		0.0081
3	0.900	0.169	0.094		0.0021
4	1.102	0.017	0.111		0.0054

Table 5. LSE results for M4 with random noises ($\sigma = 0.1$) and $t = 7$

Run	$\hat{b}_0(t)$	$\hat{b}_1(t)$	$\hat{b}_2(t)$	$\hat{b}_3(t)$	$J(t, \hat{\boldsymbol{\theta}}(t))/t$
1	0.969	0.252	0.034	0.007	0.001
2	0.902	0.235	0.078	-0.0001	0.0081
3	0.877	0.245	0.060	0.004	0.0017
4	1.080	0.091	0.077	0.004	0.0050

- **The LSE should be unbiased as the model errors for both model structures are zero mean and uncorrelated with the regressor.** (由于模型误差都是零均值且与回归量不相关，因此两种模型结构下最小二乘估计是无偏的)
- **The LSE over different runs are on average the same as the true model parameter value.** (不同次最小二乘估计平均结果与真实模型参数值相同)
- **Smaller fitting error index *J* always results in each run as the model structure M4 is larger than M3**(由于模型结构M4比M3大，其每次性能指标的拟合误差更小些)
- **However, this does not mean that the resulting parameter estimates are more accurate**(但这并不意味着所得参数估计结果更准确)

Ex.2 LSE under Effect of Model Structure Error

- **LSE Results under Non-Zero Random Noise** (非零随机噪声下的最小二乘估计结果)

Actual system with $b_0 = 1$, $b_1 = 0.1$, $b_2 = 0.1$, and $\sigma = 0.1$,
Large data set: the actual output is generated using an input sequence of $u(i) = i-1$ for $i = 1, \ldots, t = 1000$.
(采样数据集是由输入序列产生的真实输出)

Table 6. LSE results for M3 with random noises ($\sigma = 0.1$) and $t = 1000$

Run	$\hat{b}_0(t)$	$\hat{b}_1(t)$	$\hat{b}_2(t)$	$\hat{b}_3(t)$	$J(t, \hat{\boldsymbol{\theta}}(t))/t$
1	0.9955	0.1	0.1		0.0089
2	0.988	0.100	0.100		0.0106
3	1.010	0.100	0.100		0.0105
4	1.0007	0.1000	0.1000		0.0098

Table 7. LSE results for M4 with random noises ($\sigma = 0.1$) and $t = 1000$

Run	$\hat{b}_0(t)$	$\hat{b}_1(t)$	$\hat{b}_2(t)$	$\hat{b}_3(t)$	$J(t, \hat{\boldsymbol{\theta}}(t))/t$
1	1.005	0.100	0.100	0.000	0.0089
2	0.987	0.100	0.100	0.000	0.0106
3	1.012	0.100	0.100	0.000	0.0105
4	0.9939	0.1001	0.1000	0.0000	0.0098

➢ **The estimates are almost the same as their true values for each run as predicted by the theory** (如理论所预测，每次运行的估计结果与他们的真值几乎相等).

LSE under Effect of Model Structure Error

- **Summary (总结)**
  - It is important to choose the correct model structure in order to get a good fit of data and a consistent and accurate estimate of true model parameters **(选择正确的模型结构很重要，以便获得较好的数据拟合和精确且一致的真实模型参数).**
  - With a too small model structure or few model parameters to tune such as the model structures M1 and M2 in this example, it is not possible to get a good fit of the data, left alone the accurate estimate of true model parameters **(太小的模型结构或太少的可调节模型参数，如例中的M1和M2模型结构，不可能获得较好的数据拟合和精确的真实模型参数).**
  - On the other hand, with a too large model structure or too many model parameters to be determined (i.e., p is large), more complicated experimental conditions have to be met in order to guarantee the convergence of the LSE estimates **(另一方面，太大的模型结构或太多的待估计模型参数，为保证估计参数收敛会使实验条件更复杂).**
  - Furthermore, though the fit to the measured data would be better (e.g., the model structure M4), the resulting parameter estimates do not necessarily be better and the fit to another data set may be worse (overfitting or fit to the noise) **(而且，虽然对测量数据拟合更好(如M4)，参数估计结果并不一定更好,对另一数据集的拟合可能会更差(过拟合或是对噪声进行了拟合).**

Summary

- **Linear Regression Type Model Structures** (线性回归类型的模型结构)
- **Least Square Estimation (LSE) Algorithm** (最小二乘估计算法(LSE))
- **LSE under Effect of Model Structure Error** (模型结构误差对LSE的影响)